

Java Assignment 3 – Arrays

Program 1: Find Sum and Average

Write a Java program to accept n integers into an array and find:

- Sum of all elements
- Average of all elements

Example:

Input:
10 20 30 40 50

Output:
Sum = 150
Average = 30.0

Program 2: Find Largest and Smallest

Write a Java program to store n integers in an array and find the **largest** and **smallest** elements.

Example:

Input:
25 10 75 40 60

Output:
Largest = 75
Smallest = 10

Program 3: Count Even and Odd Numbers

Write a Java program to store integers in an array and count:

- Number of even elements
- Number of odd elements

Example:

Input:
10 15 20 25 30 35

Output:
Even numbers = 3
Odd numbers = 3

Program 4: Search an Element

Write a Java program to store integers in an array. Accept a number from the user and search for that number in the array.

Display the **index position** if the element is found. Otherwise, display "Element not found".

Example:

Array:
10 20 30 40 50

Search:
30

Output:
Element found at index 2

Program 5: Reverse an Array

Write a Java program to store n integers in an array and display the elements in **reverse order**.

Example:

Input:
10 20 30 40 50

Output:
50 40 30 20 10

Program 6: Find Duplicate Elements

Write a Java program to store n integers in an array and find the **duplicate elements**.

Example:

Input:
10 20 30 20 40 10 50

Output:
Duplicate elements:
10
20

Program 7: Separate Positive and Negative Numbers

Write a Java program to store n integers in an array and count and display:

- Positive numbers

- Negative numbers
- Zero values

Example:

Input:
10 -5 0 20 -8 15 0

Output:
Positive numbers: 10 20 15
Negative numbers: -5 -8
Zero values: 2

Condition: Use a single array and traverse it using a loop.

Program 8: Student Marks Using Command-Line Arguments

Write a Java program that accepts the **student name and three subject marks through command-line arguments**.

Calculate and display:

- Student name
- Total marks
- Average marks
- Result

The student is considered **PASS only if all three subject marks are 40 or above**.

Example

Compile:

```
javac StudentResultt.java
```

Run:

```
java StudentResult Ravi 80 75 90
```

Expected output:

```
Student Name: Ravi
Mark 1: 80
Mark 2: 75
Mark 3: 90
Total: 245
Average: 81.67
Result: PASS
```